

From Proof to Practice: Extracting Verified Programs to Modern C++

Bloomberg

Lean NYC meetup
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Testing is great, but not enough!

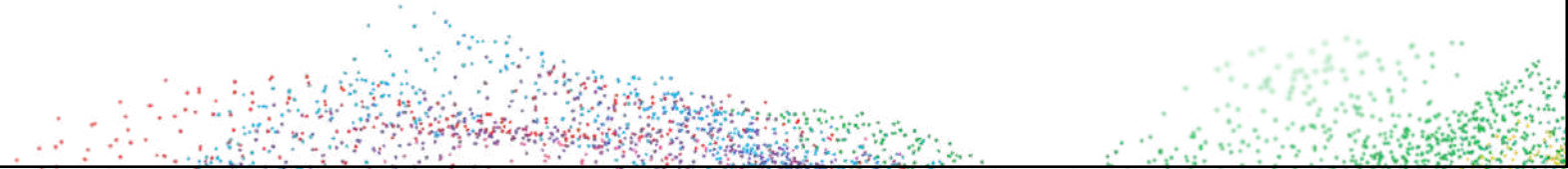
```
std::optional<std::string> parse_string(std::string_view s) {  
    if (s.empty() || s[0] != '"') return std::nullopt;  
    std::string out;  
    for (size_t i = 1; i < s.size(); i++) {  
        if (s[i] == '"') return out;  
        if (s[i] == '\\') {  
            if (++i >= s.size()) return std::nullopt;  
            switch (s[i]) {  
                case '"': out += '"'; break;  
                case '\\': out += '\\'; break;  
                case '/': out += '/'; break;  
                case 'b': out += '\b'; break;  
                case 'f': out += '\f'; break;  
                case 'n': out += '\n'; break;  
                case 'r': out += '\r'; break;  
                case 't': out += '\t'; break;  
                default: out += s[i]; break;  
            }  
        } else {  
            out += s[i];  
        }  
    }  
    return std::nullopt;  
}
```

This parser accepts `\q`, `\x`, `\a`, ..., but those are not JSON escape sequences.

Testing is great, but not enough!

Tests can find bugs, but they are **useless** for proving their **absence**.

We need **formal methods** for that!

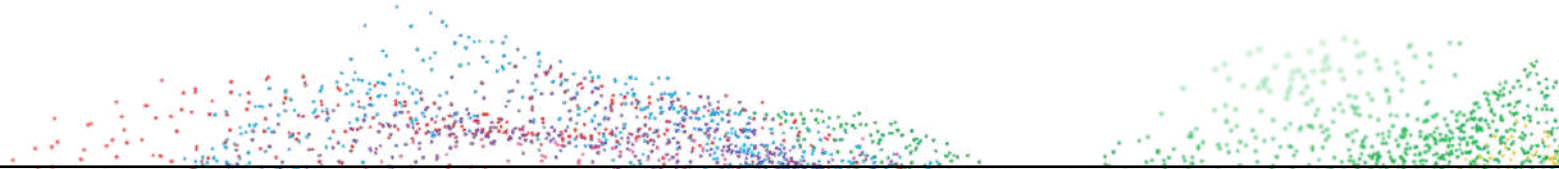


Testing is great, but not enough!

```
{  
  "role": "user",  
  "role": "admin"  
}
```

What do we do with duplicate keys?
Some parsers reject, some keep the first,
some keep the last, some keep both.

We should **specify** which.



Software verification, but which way?

Verifying existing software

- Frama-C, CBMC, VST for C
- Creusot, Verus, Aeneas, Kani for Rust
- Refinement / liquid types for Haskell, Rust, TS, Java, etc.
- Cameleer, CFML for OCaml

- Can be clunky
- Proofs are brittle
- Often not expressive enough.

?



Certified programming

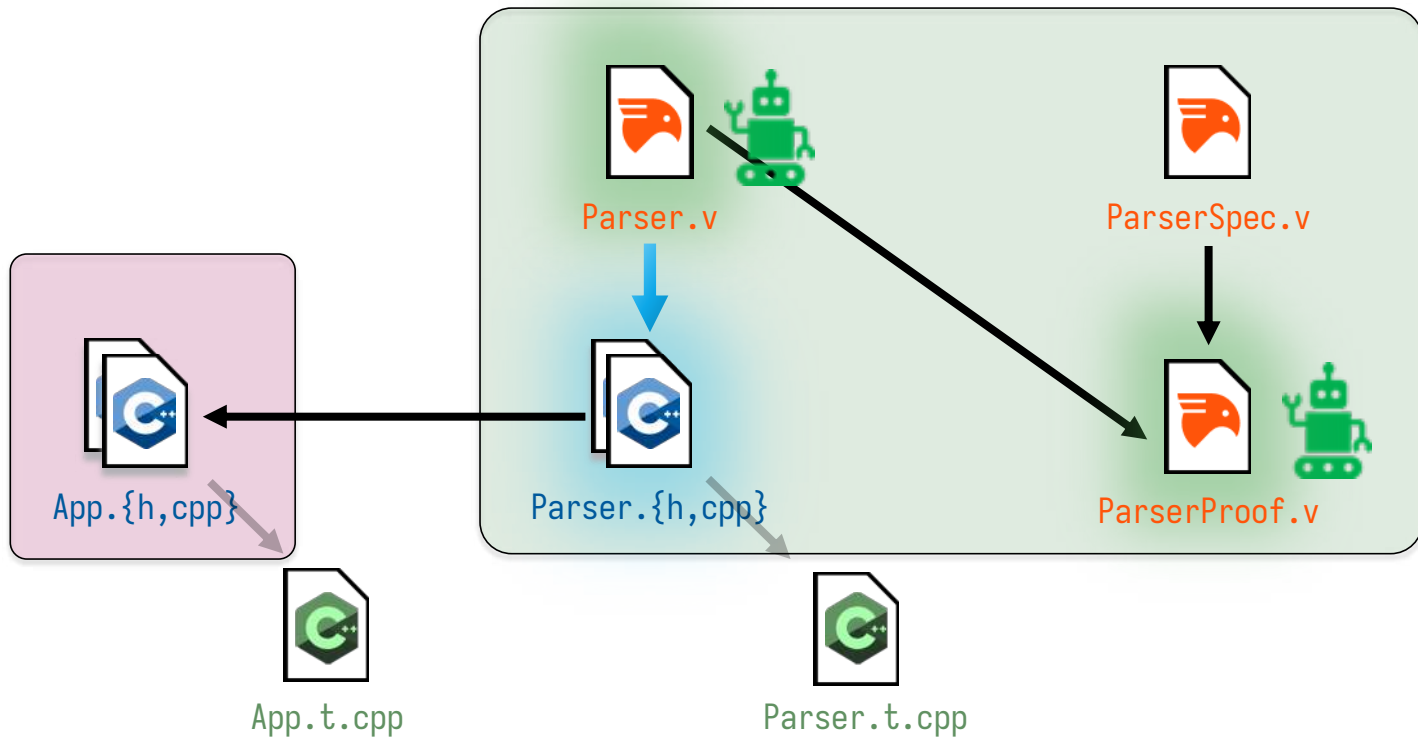
- Create new software that is **correct by construction!**
- Compile / extract the correct program into your favorite language.
- We need to make it work for Bloomberg's needs.

New Tool: Crane

- A new extraction system from  **Rocq** to **C++**.
- Generates
 - **functional-style**
 - **memory-safe***
 - **thread-safe***
 - **readable***
 - UB-free*C++ code.
- Still under active development!



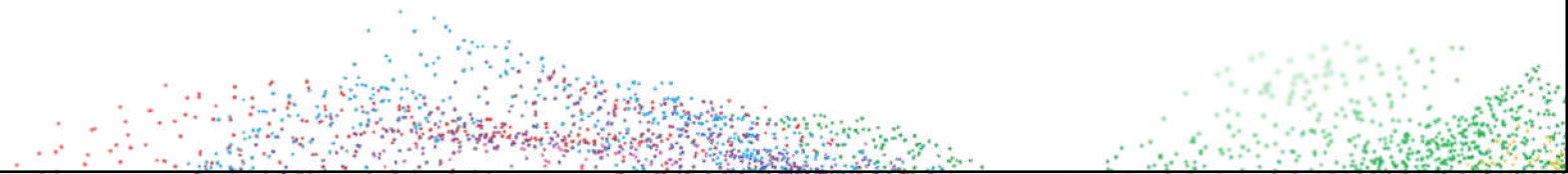
Workflow with Crane and AI agents



Why C++?

- C++ is the **lingua franca** of our engineering teams.
We are meeting them where they are.
- C++ has rich **zero-overhead abstractions**:
variant, shared_ptr, move semantics, concepts, constexpr.
- A **portable assembly language** for functional compilers!

C++'s zero-overhead abstractions turn out to be a good **compilation target** for verified functional programs.



Inductive types in functional C++

```
Inductive list (A : Type) : Type :=  
| nil : list A  
| cons : A -> list A -> list A.
```

```
template <typename A> struct List {  
    // TYPES  
    struct Nil {};  
  
    struct Cons {  
        A a;  
        std::shared_ptr<List<A>> l;  
    };  
  
    using variant_t = std::variant<Nil, Cons>;  
  
private:  
    // DATA  
    variant_t v_;  
  
    // ...  
};
```

Pattern matching in functional C++

```
Fixpoint append
  {A : Type}
  (l m : list A) : list A :=
match l with
| [] => m
| x :: l1 => x :: append l1 m
end.
```

```
template <typename A> struct List {
  // ...

  List<A> app(List<A> l2) const {
    if (std::holds_alternative<Nil>(this->v())) {
      return l2;
    } else {
      const auto &[a0, a1] = std::get<Cons>(this->v());
      return List<A>::cons(a0, a1->app(std::move(l2)));
    }
  }

  // ...
};
```

Higher-order functions in functional C++

Fixpoint map

```
{A B : Type}
(f : A -> B)
(l : list A) : list B :=
match l with
| [] => []
| x :: l' => f x :: map f l'
end.
```

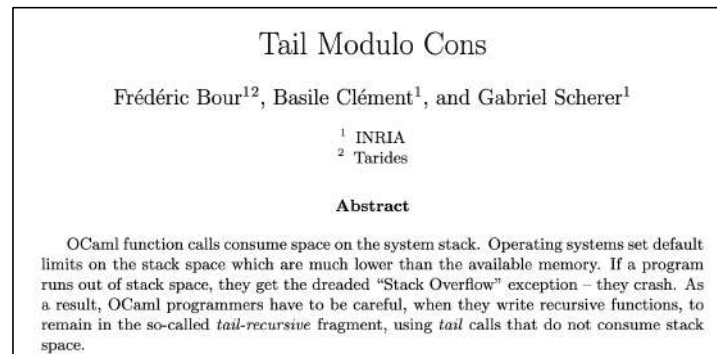
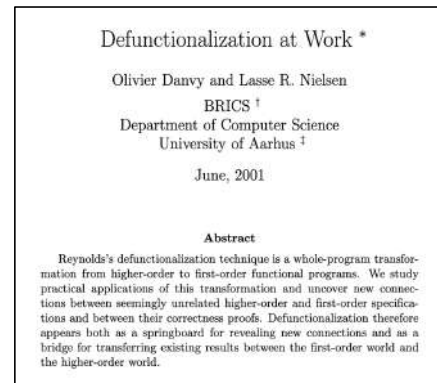
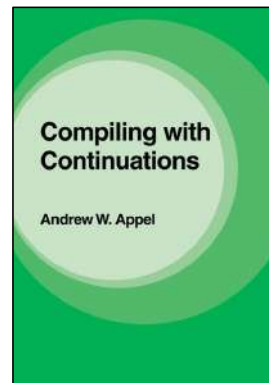
```
template <typename A> struct List {
    // ...

    template <typename B, typename F0>
        requires std::is_invocable_r_v<B, F0 &, A &>
    List<B> map(F0 &&f) const {
        if (std::holds_alternative<Nil>(this->v())) {
            return List<B>::nil();
        } else {
            const auto &[a0, a1] = std::get<Cons>(this->v());
            return List<B>::cons(f(a0), a1->map<B>(f));
        }
    }

    // ...
};
```

Loopification

- A **composition** of different transformations:
 - Continuation-passing style
 - Selective: **only** at recursive call sites!
 - Defunctionalization
 - Put frames into an explicit **stack** data structure
 - Tail call optimization
 - No stack pushing and popping
 - Tail modulo cons
 - Special but **common** case in functional programming
- The main challenge is to keep the resulting code **typeable** and **readable**.



Loopification with tail modulo cons

Fixpoint map

```
{A B : Type}
(f : A -> B)
(l : list A) : list B :=
match l with
| [] => []
| x :: l' => f x :: map f l'
end.
```

```
template <typename A> struct List {
    template <typename B, typename F0>
        requires std::is_invocable_r_v<B, F0 &, A &>
        List<B> map(F0 &&f) const {
            std::shared_ptr<List<B>> _head{};
            std::shared_ptr<List<B>> *_write = &_amp;head;
            const List *_loop_self = this;
            while (true) {
                auto &&_sv = *_loop_self;
                if (std::holds_alternative<Nil>(_sv.v())) {
                    *_write = std::make_shared(List<T1>::nil());
                    break;
                } else {
                    const auto &[a0, a1] = std::get<Cons>(_sv.v());
                    auto _cell =
                        std::make_shared(List<B>::Cons(f(a0), nullptr));
                    *_write = std::move(_cell);
                    _write = &std::get<Cons>(*_write->v_mut()).l;
                    _loop_self = a1.get();
                    continue;
                }
            }
            return std::move(*_head);
        }
};
```

Loopification with tail call optimization

```
Fixpoint is_prefix_of
  (l1 l2 : list nat) : bool :=
match l1, l2 with
| [], _ => true
| _ :: _, [] => false
| x :: xs, y :: ys =>
  if Nat.eqb x y
  then is_prefix_of xs ys
  else false
end.
```

```
bool is_prefix_of(const List<uint64_t> &l1,
                  const List<uint64_t> &l2) {
  const List<uint64_t> *_loop_l1 = &l1;
  const List<uint64_t> *_loop_l2 = &l2;
  while (true) {
    if (std::holds_alternative<Nil>(_loop_l1->v())) {
      return true;
    } else {
      const auto &[a0, a1] =
        std::get<Cons>(_loop_l1->v());
      if (std::holds_alternative<Nil>(_loop_l2->v())) {
        return false;
      } else {
        const auto &[a00, a10] =
          std::get<Cons>(_loop_l2->v());
        if (a0 == a00) {
          _loop_l1 = a1.get();
          _loop_l2 = a10.get();
        } else {
          return false;
        }
      }
    }
  }
}
```



Loopification with frames

Fixpoint length

```
{A : Type} (l : list A) : nat :=  
match l with  
| [] => 0  
| x :: xs => 1 + length xs  
end.
```

```
template <typename A> struct List {
```

```
    uint64_t length() const {  
        const List *_self = this;
```

```
    struct _Enter {  
        const List *_self;
```

```
    struct _Resume_Cons {  
        uint64_t _s0;
```

```
using _Frame = std::variant<_Enter, _Resume_Cons>;  
uint64_t _result{};
```

```
std::vector<_Frame> _stack;
```

```
_stack.reserve(8);
```

```
_stack.emplace_back(_Enter{ _self });
```

```
while (!_stack.empty()) {
```

```
    _Frame _frame = std::move(_stack.back());
```

```
    _stack.pop_back();
```

```
    if (std::holds_alternative<_Enter>(_frame)) {
```

```
        auto _f = std::move(std::get<_Enter>(_frame));
```

```
        const List *_self = _f._self;
```

```
        auto &&_sv = *_self;
```

```
        if (std::holds_alternative<Nil>(_sv.v())) {
```

```
            _result = UINT64_C(0);
```

```
        } else {
```

```
            const auto &[a0, a1] = std::get<Cons>(_sv.v());
```

```
            _stack.emplace_back(_Resume_Cons{UINT64_C(1)});
```

```
            _stack.emplace_back(_Enter{a1.get()});
```

```
        }
```

```
    } else {
```

```
        auto _f = std::move(std::get<_Resume_Cons>(_frame));
```

```
        _result = (_f._s0 + _result);
```

```
    }
```

```
    return _result;
```

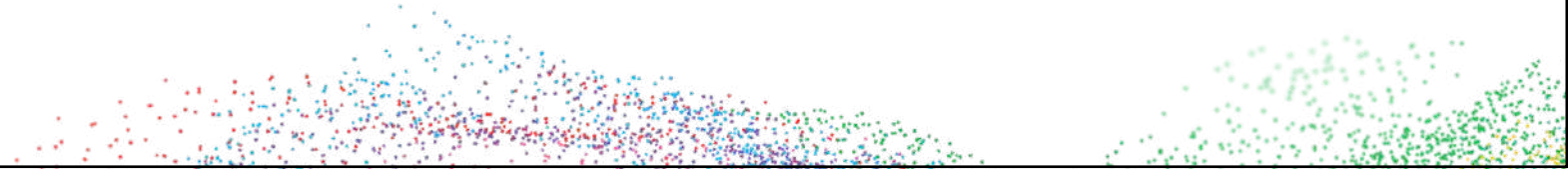
Other interesting tidbits in translating Rocq to C++

Conceptual tidbits

- Rocq **type classes** and **module types** become C++ concepts (C++20).
- Rocq **type class instances** and **modules** become C++ structs.
- Dependent types become `std::any` with casts at boundaries

Optimization tidbits

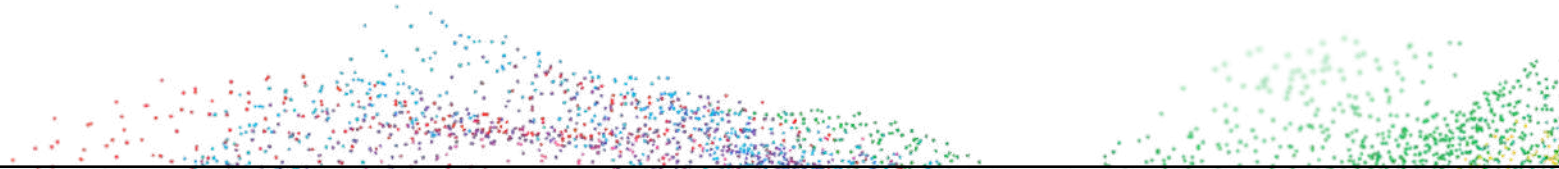
- `std::move` at **last use**.
- All-nullary inductives become `enum class`.
- Single-constructor inductives become flat structs.



Custom mappings from Rocq to C++

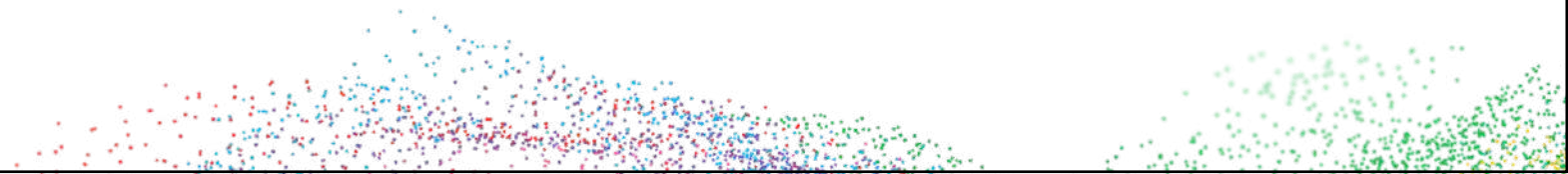
```
Inductive option (A : Type) : Type :=  
| Some : A -> option A  
| None : option A.
```

```
Crane Extract Inductive option =>  
"std::optional<%t0>"  
[ "std::make_optional<%t0>(%a0)"  
  "std::optional<%t0>()" ]  
"if (%scrut.has_value()) {  
  %t0 %b0a0 = *%scrut;  
  %br0  
} else {  
  %br1  
}"  
From "optional" "memory".
```



Expressing effectful programs

- Rocq is a **purely** functional language:
no IO, **no** state, **no** concurrency, just functions manipulating values.
- But... purely functional languages can represent effectful computation using **monads**!
- The Crane-preferred way of doing this is through **interaction trees**.



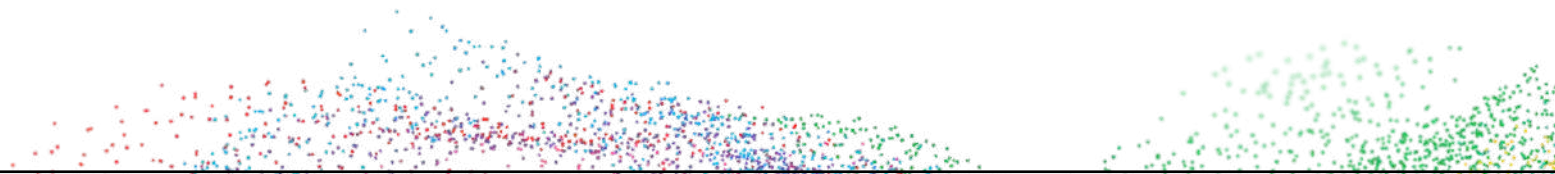
Expressing effectful programs

- The users can now write **effectful** programs in Rocq...

```
Definition test (s : string) : itree ioE unit :=  
  print ("printing " ++ s) ;;  
  ret tt.
```

- and extract them to C++!

```
void test(const std::string s) {  
  std::cout << "printing " + s;  
  return;  
}
```





<https://github.com/joom/rocqman>

A Rose Tree Is Blooming (Proof Pearl)

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Abstract

Game trees are a fundamental mathematical abstraction for analyzing games, often implemented as rose trees in functional programming. We can construct rose trees by starting from an initial state and iteratively applying the function that provides the states one move away, an approach known as an *anamorphism* (or colloquially, an *unfold*).

In ML or Haskell, finite and infinite rose trees share the same type, but proof assistants typically distinguish them: finite trees as inductive, infinite ones as coinductive. With the inductive approach, defining the unfold function is tricky but we obtain a finite tree that is easy to compute with and easy to reason about. With the coinductive approach, defining the unfold function is easy but we obtain a potentially infinite tree, which is harder to reason about since we must resort to proof techniques like bisimulation. In this paper, we compare the inductive and coinductive approaches to game trees, implement unfold functions for both, and prove the soundness and completeness of both unfold functions (with respect to the game) in the Rocq Prover. We illustrate these techniques with implementations of a tic-tac-toe game and a simple SAT solver.

CCS Concepts: • Software and its engineering → Formal software verification; Functional languages; • Computing methodologies → Game tree search.

Keywords: formal verification, termination, recursion, corecursion, game trees, Rocq

ACM Reference Format:

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1 Introduction

Game trees are a useful mathematical abstraction for reasoning about pathfinding, constraint satisfaction, and games [1]. A game tree consists of the states of a game organized as a tree, where each branch leads to a state after a move. A complete game tree is one that contains all possible game states. Such trees can be used for global reasoning about future moves, such as deterministically computing the best possible move at a given point in the game.

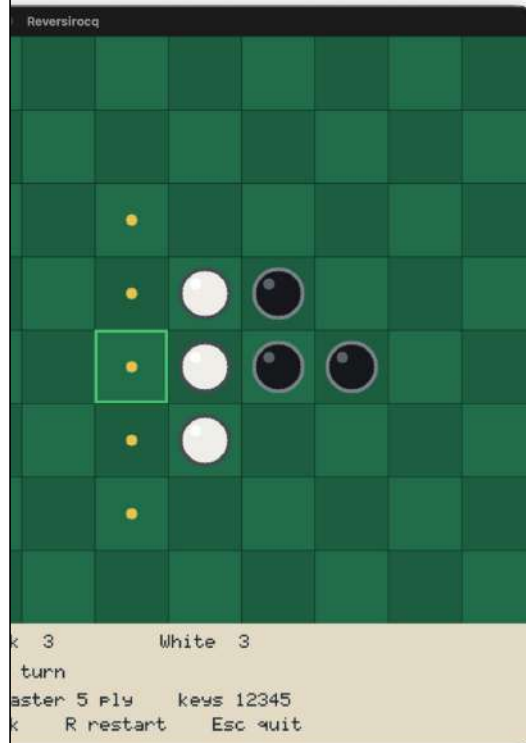


Figure 1. An example game tree for tic-tac-toe.

An example game tree for tic-tac-toe can be seen in Figure 1. At the root node, we start with the empty board, which is the initial game state. First it is x's turn, so we consider all possible moves x can make. There are 9 empty cells on the board x could choose, so the root node has 9 immediate subnodes. At each of the subnodes, it is now o's turn. There are 8 empty cells left on the board at any of these nodes, therefore they all should have 8 immediate subnodes.

We omit most branches of the tic-tac-toe game tree here for brevity. The complete tree would have 549,946 nodes if we expand naively, and that is if we were to stop building the game tree when a player wins before all cells are filled. The lesson to learn here is that game trees can grow exponentially, and that computing game trees is hard! For games with a large number of possible game states, it may even be infeasible. Famously, the chess games of IBM's Deep Blue against grandmaster and then long-time world champion Garry Kasparov exemplified the difficulty of computing game trees and searching for the best move [27, 46].

The straightforward implementation of game trees as a functional data structure coincides with what the functional programming community calls *rose trees*: trees where nodes can have a list of subtrees. This type has been the subject of considerable research, both in the functional programming community [33, 21, 23, 22, 6, 25], due to its challenges



<https://github.com/joom/reversirocq>

Writing programs with Crane

```
(** Main recursive SDL game loop. *)
CoFixpoint run_game
  (win : sdl_window) (ren : sdl_renderer)
  (ls : loop_state) : itree sdlE unit :=
  res <- process_frame ren ls ;;
  let '(quit, ls') := res in
  if quit
  then exit_game win ren
  else Tau (run_game win ren ls').
```

```
(** Program entry point. *)
Definition main : itree sdlE c_int :=
  init <- init_game ;;
  let '(win_ren, ls) := init in
  let '(win, ren) := win_ren in
  run_game win ren ls ;;
  Ret c_zero.
```

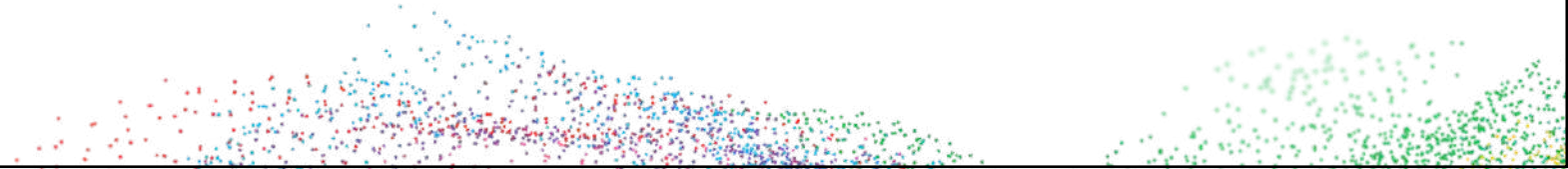
Crane Extraction "reversirocq" main.

```
// Main recursive SDL game loop.
void run_game(const sdl_window win, const sdl_renderer ren,
              const Reversirocq::loop_state &ls) {
    std::pair<bool, Reversirocq::loop_state> res =
        Reversirocq::process_frame(ren, ls);
    const bool &quit = res.first;
    const Reversirocq::loop_state &ls_ = res.second;
    if (quit) {
        exit_game(win, ren);
        return;
    } else {
        run_game(win, ren, ls_);
        return;
    }
}
```

```
// Program entry point.
int main() {
    std::pair<std::pair<sdl_window, sdl_renderer>,
              Reversirocq::loop_state> init =
        Reversirocq::init_game();
    const std::pair<sdl_window, sdl_renderer> &win_ren =
        init.first;
    const Reversirocq::loop_state &ls = init.second;
    const sdl_window &win = win_ren.first;
    const sdl_renderer &ren = win_ren.second;
    run_game(win, ren, ls);
    return 0;
}
```

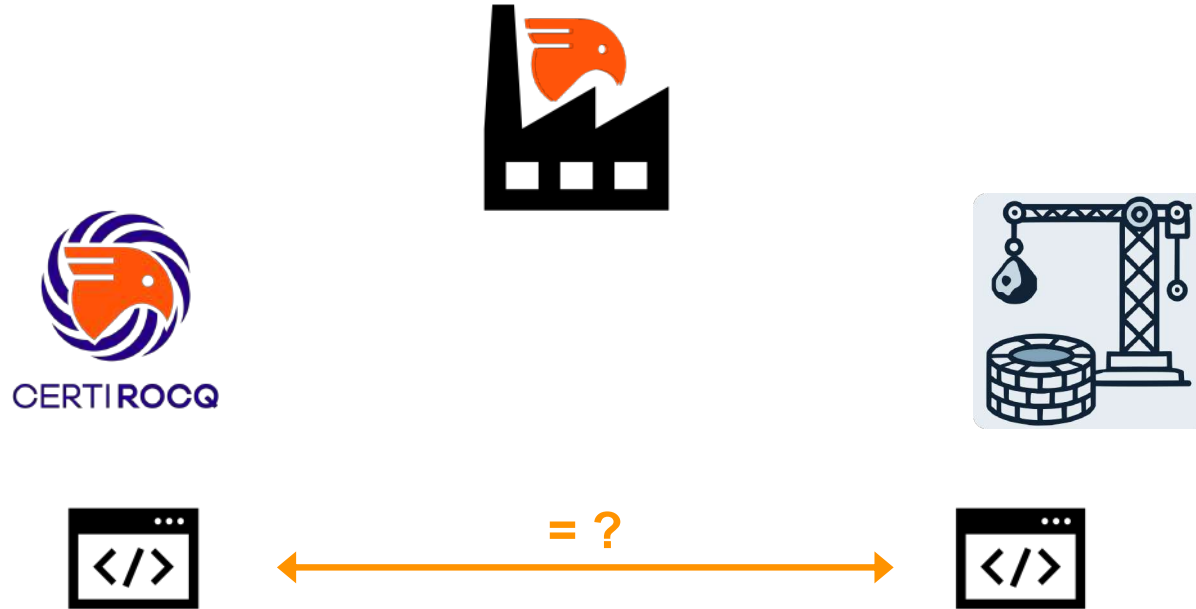

Is Crane itself verified?

- This is **lightweight formal methods**: our extractor is **not** verified.
- Generated code will also undergo the same extensive **testing** and **static analysis** all handwritten code at Bloomberg is subject to.
- We also **attack** it with AI agents and try to exploit Crane-generated code to find safety issues. We have already **found** and **fixed** some.



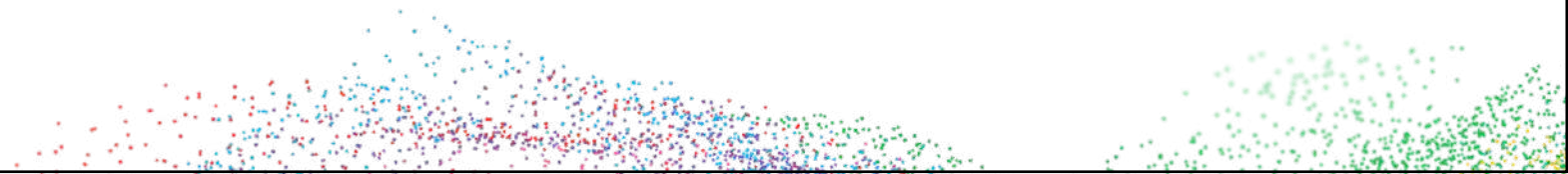
Is Crane itself verified?

- We are developing a **differential testing** framework to compare Crane and other compilers and extractors like CertiRocq.



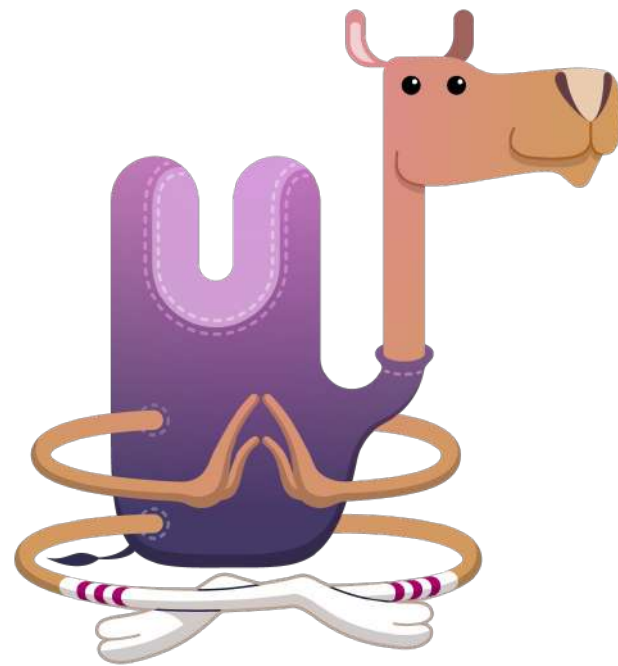
Future work

- Improve the **efficiency** of generated code without sacrificing readability *too much*.
- Extract C++ code that runs **fast**!
- Get Crane-generated code into **production**!



A note of appreciation

- Implementing **Crane** in OCaml was a **necessity** rather than a choice:
 -  **Rocq** plugins can only be written in OCaml.
- Over the last year, I got convinced that OCaml is actually a **great** choice!
 - AI agents work **faster** with a fast feedback loop!



Why Rocq is better than Lean for program verification

- Language-level issues: Coinductive types

—Coinductive types and cofixpoints exist as a library in Lean, but **not** ready for use.

```
mutual
  codata Tree α where
    | node : α → Forest α → Tree α

  codata Forest α where
    | fnil : Forest α
    | fcons : Tree α → Forest α → Forest α
end
```

```
/-- error: invalid mutual block:
either all elements of the block
must be inductive/structure declarations,
or they must all be definitions/
theorems/abbrevs -/
```

```
codata IStream α : Nat → Type where
| icons : α → IStream α (n + 1)
          → IStream α n
```

```
/-- error: Unexpected type; type will
be automatically inferred. Note that
inductive families are not supported
due to inherent limitations of QPFs -/
```

Why Rocq is better than Lean for program verification

- **Language-level issues: Nested inductive types**

—Nested inductive types and predicates are less expressive in Lean.

```
inductive Valid : Schema → JSON → Prop where
| any : Valid .any j
| str : Valid .strS (.str s)
| num : Valid .numS (.num n)
| arr : (∀ j, j ∈ elems → Valid elemSchema j) → Valid (.arrS elemSchema) (.arr elems)
| obj : Forall₂
    (fun (sf : String × Schema) (jf : String × JSON) =>
      sf.1 = jf.1 ∧ Valid sf.2 jf.2)
    schemaFields jsonFields →
    Valid (.objS schemaFields) (.obj jsonFields)

/-
-- ERROR: (kernel) invalid nested inductive datatype 'Forall₂', nested
-- inductive datatypes parameters cannot contain local variables.
-/-
```

Why Rocq is better than Lean for program verification

- **Language-level issues: Nested inductive types**

—Nested inductive types and predicates are less expressive in Lean.

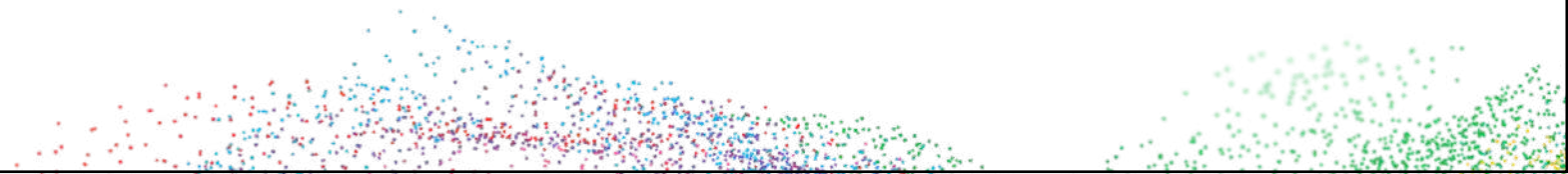
```
inductive Valid : Schema → JSON → Prop where
| ...
| obj : schemaFields.length = jsonFields.length →
    (∀ i (h1 : i < schemaFields.length) (h2 : i < jsonFields.length),
      (schemaFields[i]).1 = (jsonFields[i]).1 ∧
      Valid (schemaFields[i]).2 (jsonFields[i]).2) →
      Valid (.objS schemaFields) (.obj jsonFields)

/-
-- ERROR: (kernel) invalid nested inductive datatype 'And', nested
-- inductive datatypes parameters cannot contain local variables.
-/'
```


Why Rocq is better than Lean for program verification

- Program extraction

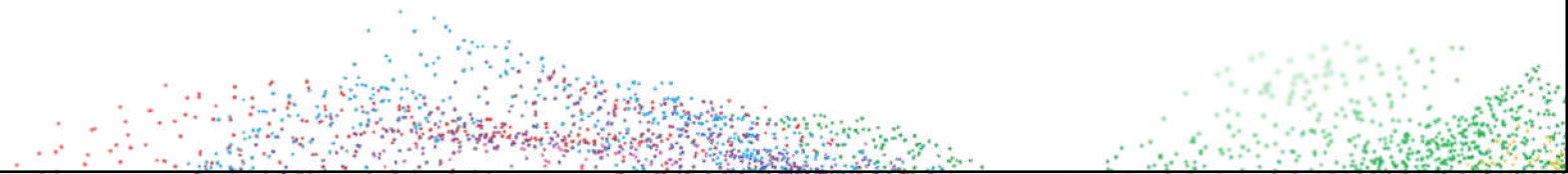
- Lean is **opinionated** in its extraction.
- Lean's extraction is **unverified** and **unspecified**.
- Rocq has **many** extraction / compilation options, with different trade-offs!
 - OCaml, Haskell, Scheme
 - a verified extraction pipeline to Malfunctor (untyped OCaml),
 - Rust,
 - Elm,
 - Clight and WebAssembly via **CertiRocq**
 - and now C++, via **Crane**



Why Rocq is better than Lean for program verification

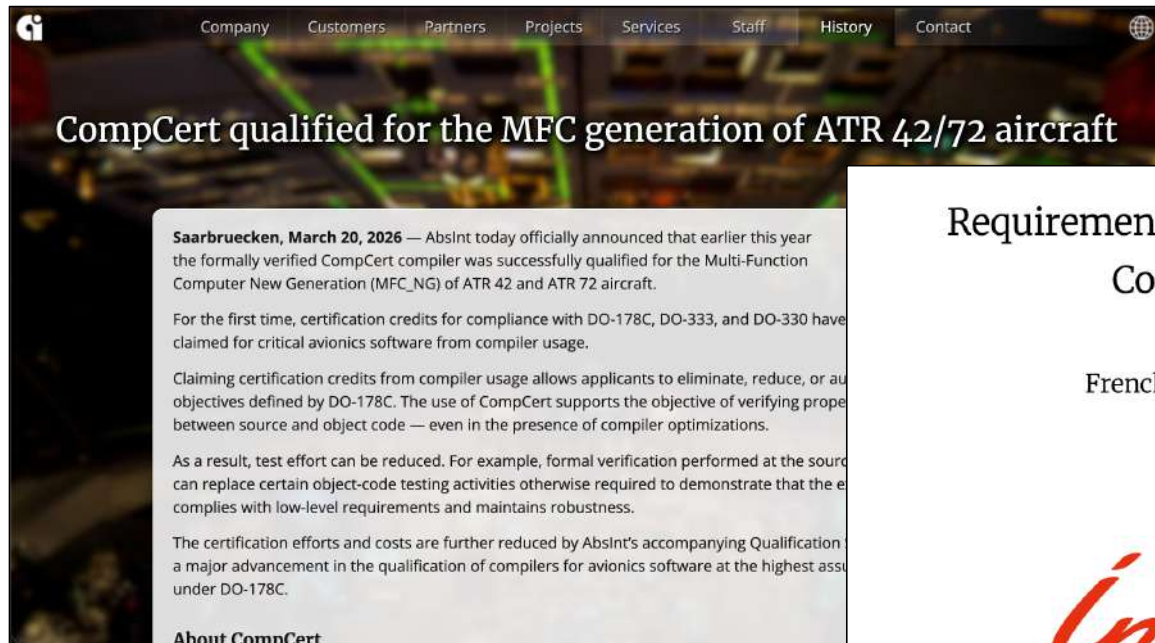
- **Missing parts in the Lean ecosystem**

- **Abstractions:** *Interaction trees, choice trees*
- **Program verification frameworks:** *Iris, VST, BRiCK, CFML*
- **Verification tools with Rocq backends/components:** *Frama-C, Why3, Cerberus*
- **Formal definitions of semantics:** *CompCert, WasmCert, JSCert*
- **Lighter-weight program verification:** *hs-to-rocq, rocq-of-ocaml, rocq-of-python, rocq-of-rust*
- **Program synthesis:** *FiatCrypto, Rupicola*
- **Verified lexing and parsing:** *Narcissus, Verbatim, CoStar*



Why Rocq is better than Lean for program verification

- Regulatory acceptance



Requirements on the Use of Coq in the Context of Common Criteria Evaluations

French National Cybersecurity Agency (ANSSI)
Inria

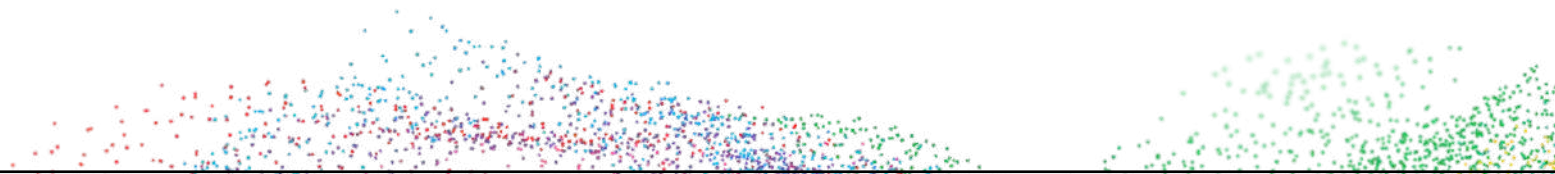
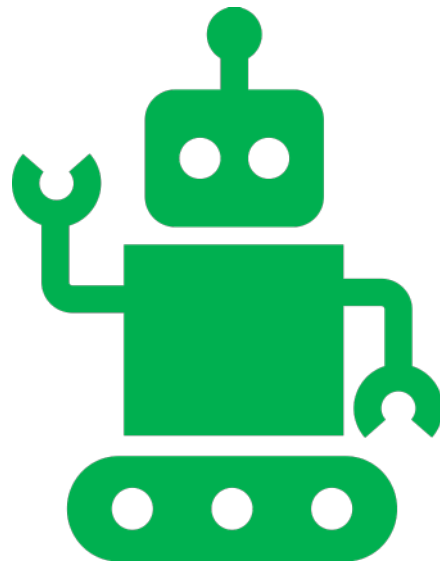
Inria



2020

Why **Rocq** is ~~better~~ than Lean for program verification as good as

- AI agents are **as good as** writing **Rocq** as they are writing Lean.
 - There is a lot of Rocq out there, written since 1989.
 - AI models do **well** now with languages they have **not** seen before.
 - Feel free to try new languages, DSLs, libraries!
AI agents can learn them and get up to speed with them.



Why Rocq is better than Lean for program verification

[–] alexk edited 4 days ago | flag | reply

▲ | I do not mean this as a dunk on QPFTypes

23

I'm the author of the QPFTypes library, and yeah, the points brought up about the rough edges are unfortunately fair points.

switching to [Lean](#). Most of this post

started life as a provocatively titled slide in a recent [LangSec](#) keynote: "Why Rocq is better than Lean for program verification."

I am not trying to start a flame v
One more caveat before I start:



Chris Henson

@_chenson_



Language-level issues

Coinductive types and co

Wojciech Różowski and Joachim
command ([implementation](#)). It is

I also want codata in Type that

alternatives use ordinary functions and structures or library encodings.

The closest Lean experiment I found to a Rocq-style codata declaration is a specification into a library encoding and generates destructor, corec examples use QPFTypes' pinned Lean 4.25.0 toolchain (latest support

Declaring codata

I do not mean this as a dunk on QPFTypes: it says "proof of concept" Rocq. For instance, Rocq accepts this parameterless coinductive type

I'm at my quota for spicy titles given yesterday's Collatz shenanigans, but this is a very nicely written and level-headed summary of what is missing in Lean compared to Rocq. (Though I hope with increased focus on program verification this changes over the next couple of years!)

ork today" everywhere below.
entum.

as the coinductive
that can be extracted.

declarations in Type; its



markusde @markusde@mathstodon.xyz

4d ...

People are verifying code in Rocq. And it keeps happening. This post is about why.

(It's quite a nice post even if the points listed do nothing for me)

Rocq

```
CoInductive co_unit : Type :=  
| co_unit_loop : co_unit.
```

Lean

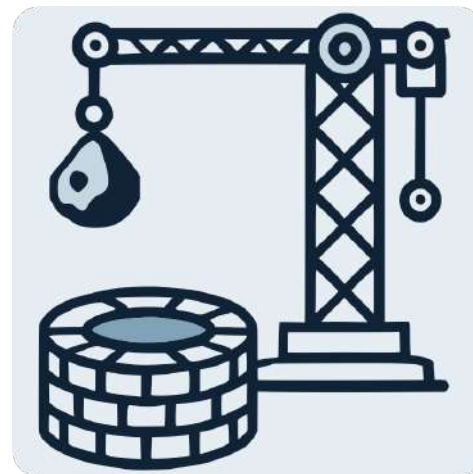
```
codata CoUnit where  
| loop : CoUnit
```

/--

error: Due to a bug, codatatypes without any parameters don't quite

Summary

- Crane is a new extraction system from **Rocq** to C++, that generates **functional-style**, **memory-safe**, **thread-safe**, **readable** C++ code.
- Using Crane, we are implementing verified libraries in C++ that will be used by Bloomberg engineers in the future.
- Formal verification now is **much cheaper** than it used to be, but we still don't have a good way of bringing verified programs to production. **We're fixing that!**



Thank you!

Check out our project at:
<https://github.com/bloomberg/crane>

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